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Control of a Spacecraft with Time-Varying Propellant Slosh Parameters

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Abstract: This paper studies the thrust vector control problem for an upper-stage rocket with fuel slosh dynamics. The control inputs are the gimbal deflection angle of a main engine and a pitching moment about the center of mass of the spacecraft. It is assumed that the rocket acceleration due to the main engine thrust is large enough so that surface tension forces do not significantly affect the propellant motion during main engine burns. The prominent sloshing modes are represented by a multi-mass-spring model with time-varying parameters. A time-varying nonlinear feedback controller is designed to control the translational velocity vector and the attitude of the spacecraft, while suppressing the sloshing modes. A simulation example is included to illustrate the effectiveness of the controller.

Keywords: Nonlinear control, propellant slosh, underactuated system.

1. INTRODUCTION

Slosh is defined as the wave motion of the free surface of the liquid inside a tank or container. It has adverse effects on the dynamics of spacecraft with liquid propellant tanks, ships and trucks transporting liquids, and robot manipulators moving liquid containers. To mitigate these effects, a variety of passive techniques have been proposed, including the use of baffles, bladders, and partitions [2, 3]. The main purpose of these techniques is to increase energy dissipation and limit the movement of liquid fuel. These techniques do not completely succeed in cancelling the sloshing effects. Moreover, these suppression methods add mass and structural complexity to the vehicle. Therefore, it is desirable to control the sloshing using active methods.

Several researchers have investigated the feasibility of applying active feedback control. The control approaches developed for robotic systems moving liquid filled containers [4, 15] and for accelerating space vehicles [1, 13, 14] are mostly based on linearized models that allow the use of linear control design methods. The literature that employ nonlinear control techniques based on the complete nonlinear dynamics formulation includes [5, 10, 11].

This paper introduces a multi-mass-spring model of sloshing fuel during a typical thrust vector control maneuver. It is assumed that the rocket acceleration due to the main engine thrust is large enough so that surface tension forces do not significantly affect the propellant motion during main engine burns. Our previous work in [11] considered a spacecraft with multiple fuel slosh modes assuming constant physical parameters. The previous results are extended to account for the time-varying nature of the slosh parameters, which renders stability analysis more difficult. The control inputs are defined by the gimbal deflection angle of a non-throttable thrust engine and a pitching moment about the center of mass of the spacecraft. The control objective is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. The results are applied to the AVUM upper stage–the fourth stage of the European launcher Vega [9]. The main contributions in this paper are (i) the development of a full nonlinear mathematical model with time-varying slosh parameters and (ii) the design of a time-varying nonlinear feedback controller. A simulation example is included to illustrate the effectiveness of the controller.

2. MATHEMATICAL MODEL

In this section, we formulate the dynamics of a spacecraft with a single propellant tank including the prominent fuel slosh modes. A Newtonian formulation is employed to express the equations of motion in terms of the spacecraft translational velocity vector, the angular velocity, and the internal coordinates representing the slosh modes. A multi-mass-spring model of the sloshing fuel with time-varying parameters is introduced to represent the prominent sloshing modes.

Following the development in [5], consider a rigid spacecraft moving on a plane as indicated in Fig. 1, where $v_x,\ v_z$ are the axial and transverse components, respectively, of the velocity of the center of the fuel tank, and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The fluid is modeled by the moment of inertia I_0 assigned to a rigidly attached mass m_0 and point masses $m_i,\ i = 1, \cdots, N$, whose relative positions along the spacecraft fixed z-axis are denoted by s_i. Moments of inertia I_i of these masses are usually negligible. The locations h_0 and h_i are referenced to the center of the tank. A restoring force $-k_i s_i$ acts on the mass m_i whenever the mass is displaced from its neutral position $s_i = 0$. A thrust F is produced by a gimballed thrust engine as shown in Fig. 1, where δ denotes the gimbal deflection angle, which is considered as one of the control inputs. A pitching moment M is also available for control purposes. The constants in the problem are the spacecraft dry mass m and moment of inertia I; the distance b between the body z-axis and the spacecraft center of mass location along the longitudinal axis, and the distance d from the gimbal pivot to the spacecraft center of mass. If the tank center is in front of the spacecraft center

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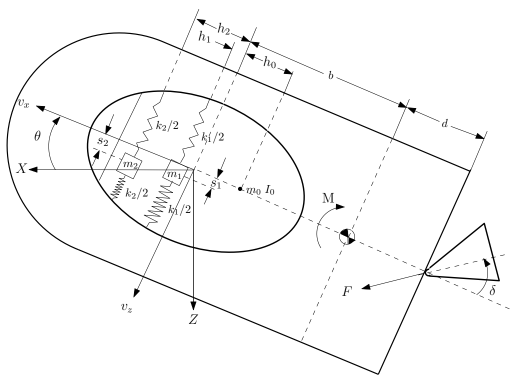

Fig. 1 A multiple slosh mass-spring model for a spacecraft.

of mass then $b > 0$. The parameters m_0, m_i, h_0, h_i, k_i and I_0 depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank. Note that these parameters are time-varying, which renders the Lyapunov-based stability analysis more difficult.

To preserve the static properties of the liquid, the sum of all the masses must be the same as the fuel mass m_f, and the center of mass of the model must be at the same elevation as that of the fuel, i.e.

$$m_0 + \sum_{i=1}^{N} m_i = m_f, \tag{1}$$

$$m_0 h_0 + \sum_{i=1}^{N} m_i h_i = 0. \tag{2}$$

Assuming a constant fuel burn rate, we have $m_f = m_{ini}(1 - t/t_f)$, where m_{ini} is the initial fuel mass in the tank and t_f is the time at which, at a constant rate, all the fuel is burned.

To compute the slosh parameters, a simple equivalent cylindrical tank is considered together with the model described in [3], which can be summarized as follows. Assuming a constant propellant density, the height of still liquid inside the cylindrical tank is $h = 4m_f/(\pi\varphi^2\rho)$, where φ and ρ denote the diameter of the tank and the propellant density, respectively.

Every slosh mode is defined by the parameters

$$m_i = m_f \left[\frac{\varphi \tanh(2\xi_i h/\varphi)}{\xi_i(\xi_i^2 - 1)\, h} \right], \tag{3}$$

$$h_i = \frac{h}{2} - \frac{\varphi}{2\xi_i} \left[\tanh(\xi_i h/\varphi) - \frac{1 - \cosh(2\xi_i h/\varphi)}{\sinh(2\xi_i h/\varphi)} \right], \tag{4}$$

$$k_i = \frac{m_i g}{\varphi} 2\xi_i \tanh(2\xi_i h/\varphi), \tag{5}$$

where ξ_i, $\forall i$, are constant parameters given by

$$\xi_1 = 1.841, \quad \xi_2 = 5.329, \quad \xi_i \simeq \xi_{i-1} + \pi,$$

and g is the axial acceleration of the spacecraft. For the rigidly attached mass, m_0 and h_0 are obtained from (1)-(4). Assuming that the liquid depth ratio for the cylindrical tank (i.e. h/φ) is less than two, the following relations apply

$$I_0 = \left(1 - 0.85\frac{h}{\varphi}\right) m_f \left(\frac{3\varphi^2}{16} + \frac{h^2}{12}\right) - m_0 h_0^2 - \sum_{i=1}^{N} m_i h_i^2 \quad \text{if } \frac{h}{\varphi} < 1,$$

$$I_0 = \left(0.35\frac{h}{\varphi} - 0.2\right) m_f \left(\frac{3\varphi^2}{16} + \frac{h^2}{12}\right) - m_0 h_0^2 - \sum_{i=1}^{N} m_i h_i^2 \quad \text{if } 1 \le \frac{h}{\varphi} < 2.$$

Let $\hat{i}$ and $\hat{k}$ be the unit vectors along the spacecraft-fixed longitudinal and transverse axes, respectively, and denote by $(x,\, z)$ the inertial position of the center of the fuel tank. The position vector of the center of mass of the vehicle can then be expressed in the spacecraft-fixed coordinate frame as

$$\vec{r} = (x - b)\hat{i} + z\hat{k}.$$

The inertial velocity and acceleration of the vehicle can be computed as

$$\dot{\vec{r}} = v_x\hat{i} + (v_z + b\dot{\theta})\hat{k},\ \ddot{\vec{r}} = (a_x + b\dot{\theta}^2)\hat{i} + (a_z + b\ddot{\theta})\hat{k},$$

where we have used the fact that

$$(v_x,\, v_z) = (\dot{x} + z\dot{\theta},\ \dot{z} - x\dot{\theta}),\ (a_x,\, a_z) = (\dot{v}_x + v_z\dot{\theta},\ \dot{v}_z - v_x\dot{\theta}).$$

Similarly, the position vectors of the fuel masses $m_0,\ m_i,\ \forall i$, in the spacecraft-fixed coordinate frame are given, respectively, by

$$\vec{r}_0 = (x + h_0)\hat{i} + z\hat{k},\ \vec{r}_i = (x + h_i)\hat{i} + (z + s_i)\hat{k},\ \forall i.$$

The inertial accelerations of the fuel masses can be computed as

$$\ddot{\vec{r}}_0 = (a_x - h_0\dot{\theta}^2 + \ddot{h}_0)\hat{i} + (a_z - 2\dot{h}_0\dot{\theta} - h_0\ddot{\theta})\hat{k},$$
$$\ddot{\vec{r}}_i = (a_x + s_i\ddot{\theta} - h_i\dot{\theta}^2 + \ddot{h}_i + 2\dot{s}_i\dot{\theta})\hat{i} + (a_z + \ddot{s}_i - h_i\ddot{\theta} - s_i\dot{\theta}^2 - 2\dot{h}_i\dot{\theta})\hat{k},\ \forall i.$$

Now Newton's second law for the whole system can be written as

$$\vec{F} = F(\hat{i}\cos\delta + \hat{k}\sin\delta) = m\ddot{\vec{r}} + \sum_{i=0}^{N} m_i \ddot{\vec{r}}_i. \tag{6}$$

The total torque with respect to the tank center can be expressed as

$$\vec{\tau} = \left(I + I_0 + \sum_{i=1}^{N} I_i\right)\ddot{\theta}\hat{j} + \vec{\rho}\times m\ddot{\vec{r}} + \sum_{i=0}^{N} \vec{\rho}_i \times m\ddot{\vec{r}}_i, \tag{7}$$

where $\vec{\tau} = [M + F(b + d)\sin\delta]\,\hat{j}$ and $\vec{\rho}$, $\vec{\rho}_0$, and $\vec{\rho}_i$ are the positions of m, m_0, and m_i relative to the tank center, respectively, i.e.

$$\vec{\rho} = -b\hat{i},\ \vec{\rho}_0 = h_0\hat{i},\ \vec{\rho}_i = h_i\hat{i} + s_i\hat{k},\ \forall i.$$

We include the dissipative effects due to fuel slosh, described by damping constants c_i. When the damping is small, it can be represented accurately by equivalent linear viscous damping. Newton's second law for the fuel mass m_i can be written as

$$m_i(\ddot{s}_i+a_z-h_i\ddot{\theta}-s_i\dot{\theta}^2-2\dot{h}_i\dot{\theta})=-c_i\dot{s}_i-k_is_i. \tag{8}$$

Using (6)-(8), the equations of motion can be obtained as

$$(m+m_f)a_x+mb\dot{\theta}^2+\sum_{i=1}^{N}m_i(s_i\ddot{\theta}+2\dot{s}_i\dot{\theta}+\ddot{h}_i) + m_0\ddot{h}_0=F\cos\delta, \tag{9}$$

$$(m+m_f)a_z+mb\ddot{\theta}+\sum_{i=1}^{N}m_i(\ddot{s}_i-s_i\dot{\theta}^2-2\dot{h}_i\dot{\theta}) - 2m_0\dot{h}_0\dot{\theta}=F\sin\delta, \tag{10}$$

$$\hat{I}\ddot{\theta}+\sum_{i=1}^{N}m_i\left[a_xs_i-h_i\ddot{s}_i+s_i\ddot{h}_i+2(s_i\dot{s}_i+h_i\dot{h}_i)\dot{\theta}\right] + 2m_0h_0\dot{h}_0\dot{\theta}+mba_z=M+Fp\sin\delta, \tag{11}$$

$$\ddot{s}_i+a_z-h_i\ddot{\theta}-s_i\dot{\theta}^2-2\dot{h}_i\dot{\theta}+\omega_i^2s_i+2\zeta_i\omega_i\dot{s}_i=0,\ \forall i, \tag{12}$$

where $\omega_i^2=k_i/m_i,\ 2\zeta_i\omega_i=c_i/m_i,\ \forall i, p=b+d$, and

$$\hat{I}=I+I_0+mb^2+m_0h_0^2+\sum_{i=1}^{N}\left[I_i+m_i(h_i^2+s_i^2)\right].$$

Here ω_i and ζ_i, $\forall i$, denote the undamped natural frequencies and damping ratios, respectively.

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the fuel slosh modes. Eqs. (9)-(12) model interesting examples of underactuated mechanical systems. The published literature on the dynamics and control of such systems includes the development of theoretical controllability and stabilizability results for a large class of systems using tools from nonlinear control theory and the development of effective nonlinear control design methodologies [12] that are applied to several practical examples, including underactuated space vehicles [7] and underactuated manipulators [8].

3. FEEDBACK CONTROLLER

This section presents a detailed development of feedback control laws through the model obtained via the multi-mass-spring analogy.

Consider the model of a spacecraft with a gimballed thrust engine shown in Fig. 1. If the thrust F during the fuel burn is a positive constant, and if the gimbal deflection angle and pitching moment are zero, $\delta=M=0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_z=\bar{v}_z,\ \theta=\bar{\theta},\ \dot{\theta}=0,\ s_i=0,\ \dot{s}_i=0,\ \forall i,$$

where $\bar{v}_z$ and $\bar{\theta}$ are arbitrary constants. Without loss of generality, the subsequent analysis considers the relative equilibrium at the origin, i.e. $\bar{v}_z=\bar{\theta}=0$. Note that the relative equilibrium corresponds to the vehicle axial velocity $v_x(t)=v_{x_0}+\bar{a}_xt,\ t\leq t_b$, where v_{x_0} is the initial axial velocity of the spacecraft, t_b is the fuel burn time, and $\bar{a}_x=F/(m+m_f)$.

Now assume the axial acceleration term a_x is not significantly affected by small gimbal deflections, pitch changes and fuel motion (an assumption verified in simulations). Consequently, Eq. (9) becomes:

$$\dot{v}_x+\dot{\theta}v_z=\bar{a}_x. \tag{13}$$

Substituting this approximation leads to the following reduced equations of motion for the transverse, pitch and slosh dynamics:

$$(m+m_f)\hat{a}_z+mb\ddot{\theta}+\sum_{i=1}^{N}m_i(\ddot{s}_i-s_i\dot{\theta}^2-2\dot{h}_i\dot{\theta}) - 2m_0\dot{h}_0\dot{\theta}=F\sin\delta, \tag{14}$$

$$\hat{I}\ddot{\theta}+\sum_{i=1}^{N}m_i\left[\bar{a}_xs_i-h_i\ddot{s}_i+s_i\ddot{h}_i+2(s_i\dot{s}_i+h_i\dot{h}_i)\dot{\theta}\right] + 2m_0h_0\dot{h}_0\dot{\theta}+mb\hat{a}_z=M+Fp\sin\delta, \tag{15}$$

$$\ddot{s}_i+\hat{a}_z-h_i\ddot{\theta}-s_i\dot{\theta}^2-2\dot{h}_i\dot{\theta}+\omega_i^2s_i+2\zeta_i\omega_i\dot{s}_i=0,\ \forall i, \tag{16}$$

where $\hat{a}_z=\dot{v}_z-\dot{\theta}v_x(t)$. Here $v_x(t)$ is considered as an exogenous input. The subsequent analysis is based on the above equations of motion for the transverse, pitch and slosh dynamics of the vehicle.

We now design a nonlinear controller to stabilize the relative equilibrium at the origin of the Eqs. (14)-(16). Our control design is based on a Lyapunov function approach. By defining control transformations from $(\delta,\ M)$ to new control inputs $(u_1,\ u_2)$, Eqs. (14) and (15) can be written as

$$\dot{v}_z=u_1+\dot{\theta}v_x(t), \tag{17}$$

$$\ddot{\theta}=u_2. \tag{18}$$

We consider the following candidate Lyapunov function to design a feedback controller for the system defined by Eqs. (17) and (18):

$$V=\frac{r_1}{2}v_z^2+\frac{r_2}{2}\theta^2+\frac{r_3}{2}\dot{\theta}^2,$$

where $r_1,\ r_2$, and r_3 are positive constants so that the function V is positive definite.

The time derivative of V along the trajectories of (17)-(18) can be computed as

$$\dot{V}=r_1v_z\dot{v}_z+r_2\theta\dot{\theta}+r_3\dot{\theta}\ddot{\theta},$$

or rewritten in terms of the new control inputs

$$\dot{V}=(r_1v_z)\,u_1+(r_1v_xv_z+r_2\theta+r_3u_2)\,\dot{\theta}.$$

Clearly, the feedback laws

$$(u_1,\, u_2) = \Big(-l_1 v_z,\, -(r_2\theta + l_2\dot{\theta})/r_3\Big), \tag{19}$$

where $l_1,\ l_2$ are positive constants and taking into account that

$$r_1 v_x v_z \dot{\theta} \leq \left(\frac{\dot{\theta}^2}{2} + \frac{(r_1 v_x v_z)^2}{2}\right),$$

yield

$$\begin{aligned}\dot{V} &= -l_1 r_1 v_z^2 - l_2\dot{\theta}^2 + r_1 v_x v_z \dot{\theta} \\ &\leq -r_1\left(l_1 - \frac{r_1 v_x^2}{2}\right) v_z^2 - \left(l_2 - \frac{1}{2}\right)\dot{\theta}^2.\end{aligned}$$

which satisfies $\dot{V} \leq 0$ if $l_1 > 0.5 r_1 v_x^2$ and $l_2 > 0.5$.

The closed-loop system for $(v_z,\, \theta)$-dynamics can be written as

$$\dot{v}_z = -l_1 v_z + \dot{\theta} v_x(t), \tag{20}$$

$$\ddot{\theta} = -K_1\theta - K_2\dot{\theta}, \tag{21}$$

where $K_1 = r_2/r_3$ and $K_2 = l_2/r_3$.

Eq. (21) can be easily solved in the case of $K_2^2 > 4K_1$ as

$$\theta\,(t) = Ae^{-\lambda_1 t} + Be^{-\lambda_2 t}, \tag{22}$$

where A, B are integration constants and $-\lambda_1$, $-\lambda_2$ are the eigenvalues of the linear system (21). Therefore, $\theta\,(t)$ and $\dot{\theta}\,(t)$ can be upper bounded as

$$|\theta(t)| \leq Ce^{-\lambda t},\ \left|\dot{\theta}(t)\right| \leq De^{-\lambda t},$$

respectively, where C, D are positive constants and $\lambda = \min(\lambda_1,\, \lambda_2)$. Now, assuming that $\lambda \neq l_1$, Eq. (20) can be integrated to obtain an upper bound for $v_z(t)$ as:

$$|v_z\,(t)| \leq \alpha e^{-\beta t},$$

where α, β are positive constants. Therefore, it can be concluded that the $(v_z,\, \theta)$-dynamics are exponentially stable under the controller (19).

We now analyze the stability of the N equations defined by (16). First we show that the system described by

$$\ddot{s}_i + 2\zeta_i\omega_i\,(t)\,\dot{s}_i + \omega_i^2\,(t)\,s_i = 0, \tag{23}$$

is exponentially stable.

From Eqs. (3) and (5)

$$\omega_i\,(t) = \sqrt{\frac{2g\xi_i}{\varphi}\tanh\left(\frac{2\xi_i h\,(t)}{\varphi}\right)} \in C^1.$$

The following properties can be shown to hold:

$$\omega_i^2\,(t) \geq \varepsilon_1^2,\ p\,(t) = \frac{1}{2}\frac{\dot{\omega}_i\,(t)}{\omega_i\,(t)} + 2\zeta_i\omega_i\,(t) \geq \varepsilon_2^2,$$

$$|2\zeta_i\omega_i\,(t)| \leq 2\zeta_i\sqrt{\frac{2g\xi_i}{\varphi}} = M_1,\ \left|\omega_i^2\,(t)\right| \leq \frac{2g\xi_i}{\varphi} = M_2,$$

$$|2\dot{\omega}_i\,(t)\,\omega_i\,(t)| \leq g\left(\frac{2\xi_i}{\varphi}\right)^2 = M_3,$$

where ε_1 and ε_2 are small positive parameters given the fact that the tank will never be totally empty, but a small amount of fuel will always remain inside. For this same reason $h\,(t) > 0,\ \forall t$. Therefore, by Corollary 1 in the Appendix, the system (23) is exponentially stable.

We now write Eq. (16) as

$$\dot{\eta} = (A_1\,(t) + A_2\,(t))\,\eta + H(t), \tag{24}$$

where $\eta = [s_i,\, \dot{s}_i]^T$ and

$$A_1(t) = \begin{bmatrix} 0 & 1 \\ -\omega_i^2(t) & -2\zeta_i\omega_i(t) \end{bmatrix},\ A_2(t) = \begin{bmatrix} 0 & 0 \\ \dot{\theta}^2(t) & 0 \end{bmatrix},$$

$$H(t) = \begin{bmatrix} 0 \\ -\hat{a}_z(t) + h_i(t)\ddot{\theta}(t) + 2\dot{h}_i(t)\dot{\theta}(t) \end{bmatrix}.$$

Under the stated assumptions, $A_1(t)$ is exponentially stable and there exist positive constants λ_0, λ_1 and λ_2 such that

$$\int_0^\infty \|A_2(t)\|\,dt \leq \lambda_0,\ \ \|H(t)\| \leq \lambda_1 e^{-\lambda_2 t},\ \forall t \geq 0.$$

Hence, by Lemma 1 in the Appendix, for any initial condition the state of the system (14)-(16) converges exponentially to zero.

4. SIMULATION

The feedback control law developed in the previous section is implemented here for the fourth stage of the European launcher Vega. The first two slosh modes are included to demonstrate the effectiveness of the controller (19) by applying to the complete nonlinear system (9)-(12). The physical parameters used in the simulations are given in Table 1.

Table 1 Physical parameters for AVUM stage of Vega.

Parameter	Value	Parameter	Value
T	2.45 kN	b	-0.6 m
m	975 kg	d	1.2 m
m_{ini}	580 kg	φ	1 m
I	400 $kg \cdot m^2$	t_b	650 s
I_1	10 $kg \cdot m^2$	t_f	667 s
I_2	1 $kg \cdot m^2$	ρ	1180 kg/m^3

We consider stabilization of the spacecraft in orbital transfer, suppressing the transverse and pitching motion of the spacecraft and sloshing of fuel wile the spacecraft is accelerating. In other words, the control objective is to stabilize the relative equilibrium corresponding to a specific spacecraft axial acceleration and $v_z = \theta = \dot{\theta} = s_i = \dot{s}_i = 0, \quad i = 1, 2.$

Time responses shown in Figs. 2-4 correspond to the initial conditions $v_{x_0} = 3000\,m/s$, $v_{z_0} = 100\,m/s$, $\theta_0 = 5°$, $\dot{\theta}_0 = 0$, $s_{1_0} = 0.1\,m$, $s_{2_0} = -0.1\,m$, and $\dot{s}_{1_0} = \dot{s}_{2_0} = 0$. We assume a fuel burn time of 650 seconds. As can be seen, the transverse velocity, attitude angle, and the slosh states converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$

tends asymptotically to $F/(m+m_f)$. Note that there is a trade-off between good responses for the directly actuated degrees of freedom (the transverse and pitch dynamics) and good responses for the internal degrees of freedom (the slosh dynamics); the controller given by (19) with parameters $r_1 = 8 \times 10^{-7}$, $r_2 = 10^3$, $r_3 = 500$, $l_1 = 10^4$, $l_2 = 4 \times 10^4$ represents one example of this balance.

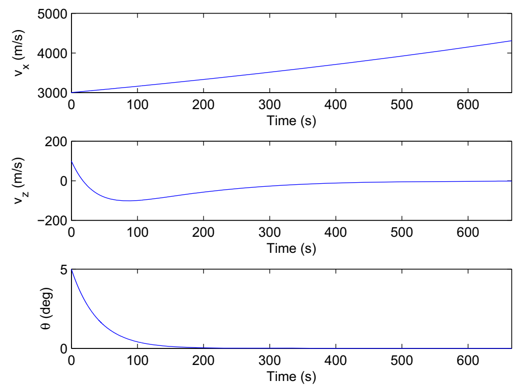

Fig. 2 Time responses of v_x, v_z and θ.

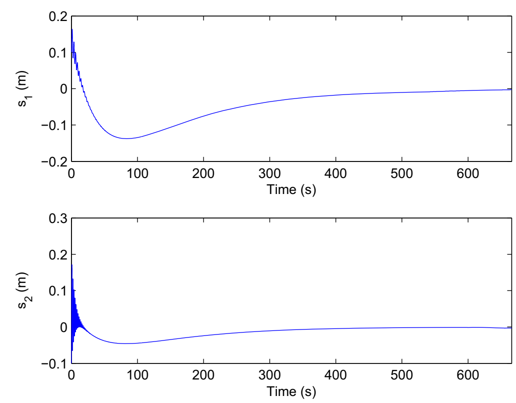

Fig. 3 Time responses of s_1 and s_2.

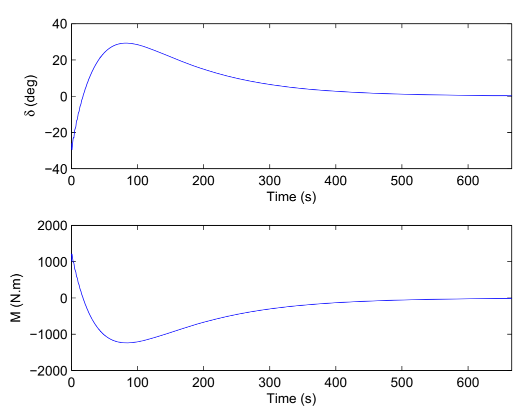

Fig. 4 Gimbal deflection angle δ and pitching moment M.

5. CONCLUSIONS

We have developed a complete nonlinear dynamical model for a spacecraft with multiple slosh modes that have time-varying parameters. We have designed a feedback control law that achieves stabilization of the pitch and transverse dynamics as well as suppression of the slosh modes, while the spacecraft accelerates in the axial direction. The effectiveness of the feedback law has been illustrated through a simulation example. The many avenues considered for future research include three dimensional maneuvers.

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APPENDIX

Consider the system

$$\ddot{s} + f(t)\,\dot{s} + g(t)\,s = 0, \tag{25}$$

where $g(t) \in C^1$, $|f(t)| < M_1$, $|g(t)| < M_2$, $|\dot{g}(t)| < M_3$.

Theorem 1: If $g(t) > \varepsilon_1^2$ and $p(t) = \frac{1}{2}\frac{\dot{g}(t)}{g(t)} + f(t) > \varepsilon_2^2$, then the origin is globally uniformly asymptotically stable.

Proof: Given the conditions above, the following bounds can be set

$$-M_1 < f(t) < M_1,\ \alpha_1^2 < g(t) < M_2,$$
$$-M_3 < \dot{g}(t) < M_3,\ \varepsilon_2^2 < p(t) < \frac{M_3}{2\varepsilon_1^2} + M_1.$$

Consider the following candidate Lyapunov function

$$V(\eta,t) = \frac{1}{2}\left(s^2 + 2\beta\frac{s\dot{s}}{\sqrt{g(t)}} + \frac{\dot{s}^2}{g(t)}\right),$$

where $\eta = [s,\ \dot{s}]^T$ is the state vector and β is a positive constant. This function can be written in a matrix form as

$$V(\eta,t) = \frac{1}{2}\begin{bmatrix} s & \dot{s} \end{bmatrix}\begin{bmatrix} 1 & \frac{\beta}{\sqrt{g}} \\ \frac{\beta}{\sqrt{g}} & \frac{1}{g} \end{bmatrix}\begin{bmatrix} s \\ \dot{s} \end{bmatrix},$$

which is positive definite if $\beta < 1$.

The positive definite quadratic function $\eta^T P \eta$ satisfies $\lambda_{\min}(P)\,\eta^T\eta \le \eta^T P \eta \le \lambda_{\max}(P)\,\eta^T\eta$, where $\lambda_{\min}(P)$ and $\lambda_{\max}(P)$ are the minimum and maximum eigenvalues of P given by

$$\lambda_{\min}(P) = \frac{1+g}{2g}\left[1 - \sqrt{1 - 4g\frac{1-\beta^2}{(1+g)^2}}\right],$$

$$\lambda_{\max}(P) = \frac{1+g}{2g}\left[1 + \sqrt{1 - 4g\frac{1-\beta^2}{(1+g)^2}}\right],$$

and thus we have $\gamma_1\|\eta\|^2 \le V \le \gamma_2\|\eta\|^2$, where γ_1 and γ_2 are positive constants.

Taking the time derivative of $V(\eta,\, t)$ yields

$$\dot{V} = -\frac{\beta}{\sqrt{g\,(t)}}\left[g\,(t)\,s^2 + p\,(t)\,s\dot{s} + \left(\frac{p\,(t)}{\beta\sqrt{g\,(t)}} - 1\right)\dot{s}^2\right],$$

which can be rewritten as

$$\dot{V} = -\frac{\beta}{\sqrt{g}}\left[\begin{array}{cc} s & \dot{s} \end{array}\right]\left[\begin{array}{cc} g & \frac{p}{2} \\ \frac{p}{2} & \frac{p}{\beta\sqrt{g}} - 1 \end{array}\right]\left[\begin{array}{c} s \\ \dot{s} \end{array}\right] < 0.$$

Clearly, $\dot{V} < 0$ if

$$\beta < \frac{16\varepsilon_1^5\varepsilon_2^2}{16M_2\varepsilon_1^4 + \left(M_3 + 2\varepsilon_1^2 M_1\right)^2}.$$

Note that $\dot{V}$ satisfies

$$\dot{V} \leq -\frac{\beta}{\sqrt{g}}\lambda_{\min}\left(Q\right)\left\|\eta\right\|^2.$$

It can be shown that if

$$\beta < \min\left\{1,\ \frac{\varepsilon_2^2}{(1 - M_2)\sqrt{M_2}},\frac{16\varepsilon_1^5\varepsilon_2^2}{16M_2\varepsilon_1^4 + \left(M_3 + 2\varepsilon_1^2 M_1\right)^2}\right\},$$

then, using Theorem 4.10 of [6], it can be concluded that the origin is exponentially stable. Hence, the following result can be stated.

Corollary 1: There exist $\alpha,\ \beta > 0$ such that

$$|s| < \beta e^{-\alpha(t-t_0)}, \quad |\dot{s}| < \beta e^{-\alpha(t-t_0)},\ \forall t \geq t_0.$$

The following result is a modified version of that presented in [12].

Lemma 1: Consider a system which is described by the linear time-varying differential equation

$$\dot{\eta} = \left(A_1\,(t) + A_2\,(t)\right)\eta + H\,(t)\,,\ \ \eta \in R^n. \tag{26}$$

If the matrix $A_1\,(t)$ is exponentially stable and there exist positive constants λ_0, λ_1 and λ_2 such that

(i) $\int_0^\infty \|A_2\,(t)\|dt \leq \lambda_0$, (ii) $\|H\,(t)\| \leq \lambda_1 e^{-\lambda_2 t},\ \forall t \geq 0,$

then all the solutions of (26) approach zero exponentially as t goes to ∞.

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